How Direct-Drive Electro-Optical Interfaces are Changing the Game for 800G and Beyond

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The integration of optical technology in data centers interfaces has been of growing interest as growing data traffic, the push for low-latency networks, and the need for efficient resource aggregation pushes datacenter traffic to the limit. Optical interconnects are emerging as key players in data transport at both board and chip levels, offering solutions to current data transfer limitations and boosting processing speeds with the potential of significant power savings. This technical bulletin article focuses on benefits of implementing Linear Direct Drive (LDD) optical interfaces, the need for electrical-optical co-simulation methodologies for high-speed data interconnects, and results from silicon demonstrations between Synopsys 112G Ethernet PHY IP and OpenLight's optical engine.

Why do data centers use so much energy?

In today's digital era, routine online activities such as movie streaming, video uploads, and online banking are just the tip of the iceberg when it comes to data center energy consumption. The introduction of sophisticated platforms like ChatGPT has drastically altered this landscape. The modern digital domain is now shaped by the heavy power demands of cloud computing, artificial intelligence, the rapid expansion of 5G networks, autonomous vehicles, and cryptocurrency mining. Over the last 12 years, while the total system bandwidth has expanded by a factor of 80, the total system power has only risen by a factor of 22. A notable trend is the disproportionately rapid increase in SerDes power, both in the host and in pluggable optics modules, compared to other system components. Moving the optics closer to the switch chip offers a shorter, lower loss electrical channel and further power savings. In this context, the importance of co-packaged optics (CPO) and linear pluggable optics (LPO) becomes increasingly crucial. According to Lightcounting, LPO/CPO ports are projected to constitute over 30% of the total 800G and 1.6T ports installed between 2026 and 2028.

3 Main Trends Driving Optics into Datacenters

#1 AI has created an avalanche of growth in data traffic. The surge in both business and consumer traffic is driving a substantial increase in network traffic. Dell'Oro Group predicts that by 2027, accelerated servers will make up almost 50% of the server market. Optical interconnects directly connecting GPUs at the shelf level are emerging as a solution, reducing both power usage and latency.

#2 Flattening of networks for low latency. There is a strategic shift towards flattened network architectures. By simplifying the network hierarchy, flattening significantly reduces the number of hops between data points, thereby lowering latency.

#3 Aggregation of homogeneous resources in rack. Standardizing computational resources within racks calls for resource aggregation increasingly dependent on optical interconnects. Interconnects are estimated to account for approximately 25-30% of a data center's total power consumption.

Co-Packaged Optics (CPO) & Linear Drive Pluggables (LPO) Improve Network Power Efficiency

Optical interconnects are increasingly crucial as data rates approach 224 Gbps, where copper's effectiveness diminishes. The industry is moving towards optical interconnects to reduce latency and signal integrity issues. Efforts are underway to lower the power consumption of pluggable optics from the current 8 pJ/bit to 6 pJ/bit with onboard optics, aiming for a 25% reduction. Co-packaged optics further promise to enhance power efficiency aiming for 3 pJ/bit. A critical challenge is the heat sensitivity of lasers in co-packaged optics: as electrical components heat up, they can raise the ambient temperature, potentially surpassing the optimal operating range of around 70 degrees Celsius for lasers.

What is Linear Direct Drive?

Linear Drive (Direct Drive) represents a significant evolution in optical interconnect technology. The DSP typically found on pluggable optics is eliminated; instead, the switch ASIC's PHY directly drives an optical engine on a pluggable module. This optical engine does not include retimers or DSPs but is equipped with linear amplifiers. By removing the DSP from the pluggable optics, Linear Drive systems can achieve up to a 25% decrease in power usage, while maintaining pluggability. Challenges include: more complex SerDes to enable direct drive; managing signal integrity without a dedicated DSP; heat management; and ensuring compatibility between the switch ASIC's PHY and the optical module, often requiring co-simulation and co-design.

Electronic-Photonic Co-design

Traditionally, electronic chip designers used EDA tools to model photonic components through electrical equivalents. This approach faces limitations at high data rates like PCIe 6.0 and 224G SerDes. In complex photonic circuits, factors like optical reflections, crosstalk, noise, dispersion, and nonlinearities crucially influence performance. Simulating optical signals as electrical equivalents can risk inaccurate performance estimations. A comprehensive E-O co-design approach, respecting the unique properties of both domains, is essential.

Synopsys and OpenLight are Lighting the Path for Linear Drive

Synopsys, in collaboration with OpenLight (a photonics venture formed with Juniper Networks), has demonstrated the optical eye performance of a linear electrical-optical-electrical link transceiver. The demonstration features Synopsys' 112G Ethernet PHY IP, designed for long-reach (LR) channels, driving the OpenLight photonic integrated circuit (PIC). The Ethernet PHY IP effectively compensates for over 13dB of path loss using advanced equalization schemes, operating at 106 Gbps and achieved a TDECQ of 1.46 dB. The OpenLight PIC utilizes silicon photonics technology, incorporating an integrated PIC laser and a high-speed electro-absorption modulator. The system achieved an impressive average running bit error ratio (BER) of 7.16x10^-7, showcasing exceptional end-to-end, un-retimed E-O-E link performance (demo showcased during ECOC 2023).

Summary

Optical interconnects are becoming increasingly crucial as data rates reach 800Gbps and beyond, offering solutions to the limitations of electrical interconnects in terms of latency, bandwidth, and power efficiency. Synopsys is at the forefront of linear drive optics with its silicon proven 112G and 224G Ethernet PHY IP, supported by tools like OptoCompiler for unified electronic and photonic design, OptSim for photonics simulation, and PrimeSim for electronic circuit simulation.